Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Andreas Moshovos	Email	andreas.moshovos@utoronto.ca
Institute Name	University of Toronto			
Address	10 King's College Rd, Toronto, ON, M5S 3G4, CANADA			

2. Technology Overview

Title of Proposed Technology	*Automated AI Model Quantization for Meeting Real-Time, Energy and Size at the Edge/IoT and DataCenter*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (X), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (X), AI (X), Next-gen Communication ( ), Advanced Robotics & Manufacturing (X), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Software	*Software Solutions*	R&D / Engineering Services	Other R&D / Engineering-Related Technologies
Concept of Proposed Technology	We propose to investigate and demonstrate the performance, energy-efficiency and accuracy possible with our methods for learning "optimal" datatypes (number of bits and their meaning) for the tensors of AI models. Our methods optimize any AI model so that it runs faster, cheaper, and on smaller/less powerful hardware, with little or no accuracy loss. This crucial technology delivers the following benefits depending on the AI use case: 1. *Allowing models to fit on a hardware platform whereas they couldn't run at all before.* 2. *Enabling higher performance (speed) with minimal or no effect on accuracy.* 3. *Enabling higher energy efficiency with minimal effect on accuracy.* 4. *Allowing longer context.* 5. *Using fewer nodes to run larger models.* 6. *Reducing operating costs.* 7. *Helping the model developer re-engineer the model's architecture to achieve any of the*			

	above. In more detail: We have developed methods that automatically determine what datatypes to select for each tensor (or subtensor) so that speed, energy and size objectives can be met when running a machine learning model on any hardware platform. Other methods rely on trial-and-error relying on user-directed exploration. Our method automates the process, learning "optimal" datatypes for artificial intelligence models and lossless compression methods. The project will demonstrate the benefits possible for several AI use cases in close collaboration with interested partners.
Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* ***Technology Need:*** *Today, reducing the number of bits used to represent each value in an AI model is more important than ever because model sizes, context lengths, and deployment demands are growing faster than hardware improvements. Fewer bits directly reduce memory footprint and memory bandwidth, often the primary bottlenecks in modern inference, so models can fit on smaller devices, run with lower latency, and serve more users per accelerator. Lower precision also cuts energy use and operating cost by reducing data movement, which is frequently more expensive than arithmetic. At the same time, the industry is pushing toward longer-context LLMs, multi-modal models, and on-device AI, all of which amplify memory and bandwidth pressure; improving "bits per value" is one of the most immediate and broadly applicable levers for scaling capability within fixed power, cost, and infrastructure constraints. Optimizing models to run more efficiently enables running them locally on edge/autonomous devices or on customer-controlled platforms without having to rely on cloud-services. This enhances privacy, security and enables end-users full control of their data and application deployment. This is essential especially in mission-critical applications.* ***Maturity Level:*** *We developed and validated our method across a range of openly available models spanning both LLMs and CNNs. For LLMs in particular, we quantified the speed–accuracy trade-offs using llama.cpp [https://github.com/ggml-org/llama.cpp] on a representative set of models. It is important to note that the observed trade-off curve is strongly influenced by the maturity and quality of the existing llama.cpp quantization implementations and kernels, and therefore should not be interpreted as a direct proxy for the benefits we expect on other hardware or software stacks. Working directly with potential partners would allow us to prioritize specific target hardware platforms, serving backends, and application scenarios (e.g., models other than text to text LLMs), and to generate the platform-specific measurements and proof points they need to confidently estimate end-to-end impact. Our existing measurements do not include additional benefits from customized kernel code and compression methods. Collaborating with partners is essential in selecting appropriate target platforms. Further, our technology can take advantage of specialized datasets and further optimize the models for specific use cases. Collaborating with partners will offer further opportunities for practical and relevant results.* *The lead student for the project will be Mr. Jay Ha who is Korean and has joined our group this year*

	as an MASc candidate. Further the PI will seek to hire an additional Korean student or Post Doctoral Fellow. *The group also hosts this year as a visiting professor Dr. Tae-Hwan Kim, Professor, School of Electronics and Info. Engineering, Korea Aerospace Univ., Goyang, Korea. We hope to continue collaborating after his departure.*
Key Words	1. *Optimized Artificial Intelligence Inference* 2. Automated Artificial intelligence Model Optimization for Speed, Energy and Accuracy 3. Edge and Local Inference 4. Specialized Hardware and Software for Machine Learning Inference

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project* *Investigate and demonstrate the performance, energy, and accuracy trade-offs possible by learning datatypes per tensor or subgroups of tensor values for artificial intelligence models and applications of interest.* *The research group will coordinate with potential partners on models, applications, and platforms of interest. As initial points we identify LLM and vision tasks, llama.cpp as the backend, and NVIDIA Orin class and ARM/Intel based systems for hardware.* *See Section "R&D Plan" for an example of the expected artifacts that this project will produce.* *Our methods optimizing AI models have been documented in the following publications:* • *BitPruning: Learning Bitlengths for Aggressive and Accurate Quantization* • *Schrödinger's FP: Dynamic Adaptation of Floating-Point Containers for Deep Learning Training* • *Every Bit Matters: A Hardware/Software Approach for Enabling More Powerful Machine Learning Models* • *Learning Efficient Datatypes for Deep Neural Network Training and Inference* *The current versions of our methods have been further improved and are been commercialized by ByteShape Inc (the PI and three of his former students are the co-founders).* ***Technology Need:*** *Here's a more detailed discussion as to why such methods are essential:* *Local inference is becoming important because many workloads are constrained by latency, connectivity, and cost, not just raw compute. When inference runs where data is produced, we avoid the network round trip, which can materially improve responsiveness for interactive systems and time-critical decisions. It also reduces how much data must be shipped to the cloud, which lowers bandwidth spend and can simplify deployment at scale.* *Control of data is an equally strong driver. Data sovereignty is the idea that data is governed by the laws of the region where it is generated, which pushes organizations toward architectures where sensitive inputs stay on-device or on-prem. Building on that, sovereign AI is commonly*			

	framed as maintaining jurisdictional and operational control over the AI stack, including where data and models run and who can access them. This matters for regulated sectors, national security, and enterprises that want clear governance and reduced dependency on external providers. *For robotics and autonomous systems, local inference is often a requirement rather than an optimization. Robots must react on tight control loops, and they may operate with intermittent connectivity, so relying on the cloud can introduce unacceptable delays and fragility. Recent on-device robotics efforts emphasize these points directly, highlighting robustness when networks are weak or absent and the need for low-latency on-robot execution.* *That is why model optimization is crucial. Edge and robotic platforms are constrained by memory, power, and thermal limits, so an unoptimized model may not fit, may miss real-time deadlines, or may drain the energy budget. Compression, quantization, and datatype selection reduce footprint and bandwidth demand and can move an application from "not deployable" to "reliable in the field," especially for perception and decision-making pipelines that must run continuously and safely.* *In more detail:* *Over the last decade, the key question for inference has shifted from "can we run it?" to "can we run it efficiently within strict limits on energy, memory, and latency," especially outside the data center. Power is now a primary constraint across servers, desktops, and edge devices. One of the most effective levers for reducing energy and improving speed is the datatype, meaning how many bits we use per value and how those bits represent numbers. Moving from 32-bit floating point to narrower formats can cut compute, reduce energy, and increase effective bandwidth, but poor choices can waste performance or harm accuracy.* *Datatype selection is now essential because modern inference is often limited by memory traffic, not arithmetic. A large fraction of time and energy is spent moving weights and activations through the memory hierarchy. Off-chip DRAM access is especially costly, and the most direct way to reduce it is to store and move fewer bits per value. This matters even more for edge and local inference, where bandwidth, cache size, and power budgets are tighter and there is little room for unused precision.* *At the same time, a single low-precision format for the whole model is rarely sufficient. Different tensors and layers have different numerical needs, yet today's practice still relies heavily on manual trial-and-error with a small set of formats. This often leads to using more bits than necessary in some places and too few in others. The result is either missed savings in memory, latency, and energy, or avoidable accuracy loss. Automating datatype discovery enables precision to be tailored where it matters, turning datatype choice into a reliable optimization tool.* *Better datatype choices also expand what is feasible for local and edge deployments. They can make larger models fit in device memory, reduce per-token latency for interactive applications, and lower energy enough to meet thermal or battery limits. Because the best choices depend on the specific hardware, serving stack, and workload, working with platform owners and application partners is the fastest way to produce credible, platform-specific results.*
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters* *We propose the following metrics to be measured per model and platform. For this discussion we use an LLM text-to-text agentic use-case. The benchmarks and methodology will be revised accordingly with standard benchmarks per use-case after consultation with the partners.* ***Accuracy (task success)***

	Structured correctness: schema validity plus field-level exact match for extraction/transform tasks. *Use GSM8K: Math benchmark that requires responses to be formatted in a certain way.* *Grounded Q&A: faithfulness to provided context, citation correctness (when applicable), and answer completeness.* *Use MMLU, which is a general knowledge benchmark. Caveat it is multiple choice.* *Tool-use success: correct tool selection, correct tool arguments, correct final answer after tool results.* *Use BFCL v3 which is an agentic (tool calling) benchmark.* ***Latency and throughput*** *Measure additional characteristics such as the ones below using llama.cpp.* *Decode time and ms/token (or tokens/sec).* *End-to-end latency (request start to final token).* *Tail latencies at p50/p95/p99 under each concurrency target.* ***Methodology*** *We will follow the benchmark setups as they are provided. Measurements will be taken on actual platforms using standard profiling tools and methods.* *The project will deliver measurements and reports describing the findings along with example optimized models.*
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* *The plan below assumes that interaction with potential partners to revise the workloads and use cases. If this is not possible, we will proceeding with text-to-text, image-to-text, text-to-image, and vision tasks using openly available models. Further, we will target NVIDIA Orin-class, ARM-based (e.g., Raspberry PI) and Intel-based systems (embedded class).* *The Project will proceed with considering the following workflow per model/task and target platform.* *We anticipate that iterating through each model/task will require at least 6-8 months. Accordingly over the period of 3 years we can afford to investigate 4-12 different use cases (workload/platform).* *We expect to meet with our partners regularly as this will provide us with the opportunity to better focus on applications and platforms that are the most relevant. However, we can pursue this project independently by selecting representative applications for various domains such as speech to text, text-to-text, vision and agentic tasks.* ***Per Use Case/Workload*** *Phase 1: Target workload definition* *Phase 2: HW/SW Platform and Benchmarking Definition, potential Dataset Development* *Phase 3: Bottleneck Identification* *Phase 4: Model Optimization* *Phase 5: Iterate* *Phase 5.1: Partner Validation* *Phase 5.2: Bottleneck Identification → Investigate Solutions* *Interaction with Quantization, Sparsity, Data Compression, etc.* *Phase 6: Revision of methods based on experience* ***Program goal:*** *Over three years, the project will validate a rigorous methodology for optimizing AI models for edge and local inference across multiple real-world workloads. For each use case, we will quantify and navigate trade-offs among throughput, latency and tail latency, accuracy,*

memory footprint, and energy. The project's external deliverables will be results and proof points, and in some cases optimized model artifacts.

Standard per-use-case workflow (used throughout)

- ***Phase 1: Target workload definition***
  Define task, operating conditions, accuracy requirements, and constraints on latency, memory, and energy.
- ***Phase 2: HW/SW platform and benchmarking definition; potential dataset development***
  Select target hardware and software stack, define metrics and benchmark protocol, and develop or adapt datasets where needed.
- ***Phase 3: Bottleneck identification***
  Profile end-to-end to identify dominant sources of cost.
- ***Phase 4: Model optimization***
  Produce multiple Pareto-optimal model variants using quantization and datatype selection, sparsity, and compression where appropriate.
- ***Phase 5: Iterate***
  - ***Phase 5.1: Partner validation***
    Re-run measurements on partner-relevant platforms and scenarios, confirm benefits, and capture constraints.
  - ***Phase 5.2: Bottleneck identification → investigate solutions***
    Re-profile under partner conditions and investigate improvements, including interactions among quantization, sparsity, and compression.
- ***Phase 6: Revision of methods based on experience***
  Update the internal methodology and prioritization based on what worked, what did not, and what generalizes.

Here's an overview of the proposed research plan:

Year	*Platforms*	*Use cases*	*Partner validations*	*Primary activities and deliverables (per use case follows Phases 1-6)*
1	*1-2*	*1-2*	*0*	*Define target workload(s) and success metrics (accuracy, latency, tail latency, energy).* *Set platform baseline(s) and benchmark protocol.* *Identify bottlenecks and produce initial optimized model variants.* *Deliver a concise report detailing findings and Pareto trade-offs.*
2	*1-3*	*3-4*	*2*	*Expand to additional workloads while keeping platforms focused.* *Run two partner validations to confirm results under partner constraints.* *Iterate on tail-latency and energy bottlenecks.* *Deliver updated report detailing findings and refined optimized model variants.*
3	*3*	*4-6*	*4*	*Complete a broader portfolio across edge and local inference use cases.* *Robustness and stress testing (worst-case inputs, sustained runs, concurrency where relevant).* *Run four partner validations across scenarios and platforms.* *Deliver final report and models.*

EXAMPLES OF ANALYSIS AND MODELS

Here's an example highlighting the type of analysis and optimizations that is to be expected for each model. The excerpt below is a high-level description of the accuracy-performance trade-offs that our technologies automate for two small LLMs. Similar results for a 30B class model can be found here [https://byteshape.com/blogs/Qwen3-30B-A3B-Instruct-2507/]

GGUF-Quantized Qwen3 4B Instruct 2507.

The study below considers variants of the above models which have been quantized to an average bits per weight of 5

bits down to 2.7 bits. This study showcases accuracy-size-performance tradeoffs. The advantage becomes especially clear at lower bitlengths, where our method preserves quality far better than established approaches. The models that were optimized with our methods are shown as "ByteShape".

The results have been collected on the llama.cpp backend to deliver high-performance quantized models that a wide audience can easily use. The study reports multiple quality-cost tradeoffs, plus detailed evaluations so you know exactly what to expect from each variant. For comparison, the study also considers models Unsloth, the leading llama.cpp-based quantization method, and across several established tasks.

Further results and discussion can be found here: https://byteshape.com/blogs/Qwen3-4B-I-2507/ The models are available here: https://huggingface.co/byteshape

Executive Summary:

*Compared to leading quantized models with **almost identical TPS with llama.cpp**, our **Qwen 4B Instruct 2507** variants greatly **reduce error rates**:*

- ***3.34 bpw: 2.46× less error on Raspberry Pi 5***
- ***3.55 bpw: 3.5× less error on RTX5090.***

Read on for head-to-head comparisons with leading quantization methods and for more models that explore different quality-size-speed sweet spots.

Qwen3 4B Instruct 2507

*Let's start with **Qwen3 4B Instruct 2507**. We use ShapeLearn to pick the best per-tensor GGUF weight formats to minimize model size while keeping output quality high. We focus on the **2.5–5 bits/weight** range, where quality starts to drop, and the most interesting tradeoffs appear.*

*For reference, we compare against **Unsloth-produced GGUF models**, which offer a strong set of high-quality quantizations across many models.*

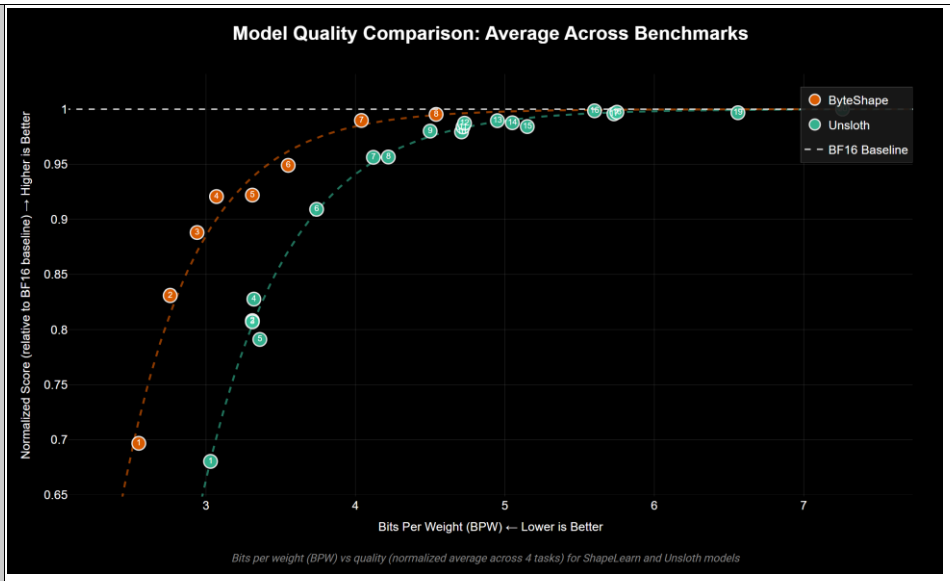
Size vs. Accuracy: Bits/Weight vs. Score

*The figure below shows **bits/weight (BPW)** vs **quality (normalized average across 4 tasks)** for ShapeLearn and Unsloth, along with their trendlines. Our models sit **above** (higher quality) and **to the left** (smaller footprint) of the Unsloth ones.*

For example, our model IQ-3.07bpw [#4 on the graph]:

- *is **18% smaller (0.67 bits)** than unsloth/Qwen3-4B-Instruct-2507-Q3_K_S [#6 on the graph] and **higher quality** with 1.15x smaller error rate(7.94% vs 9.10%),*
- *delivers **≈4.02x lower error rate** (7.94% vs 31.97%) than unsloth/Qwen3-4B-Instruct-2507-UD-IQ2_M [#1 on the graph] with only a 1% larger footprint (0.04 bits).*

*All of these points on the curve come from changing **a single ShapeLearn** (our datatype learning method) **hyperparameter** that controls quantization aggressiveness, which lets us smoothly move along the quality–cost trade-off.*

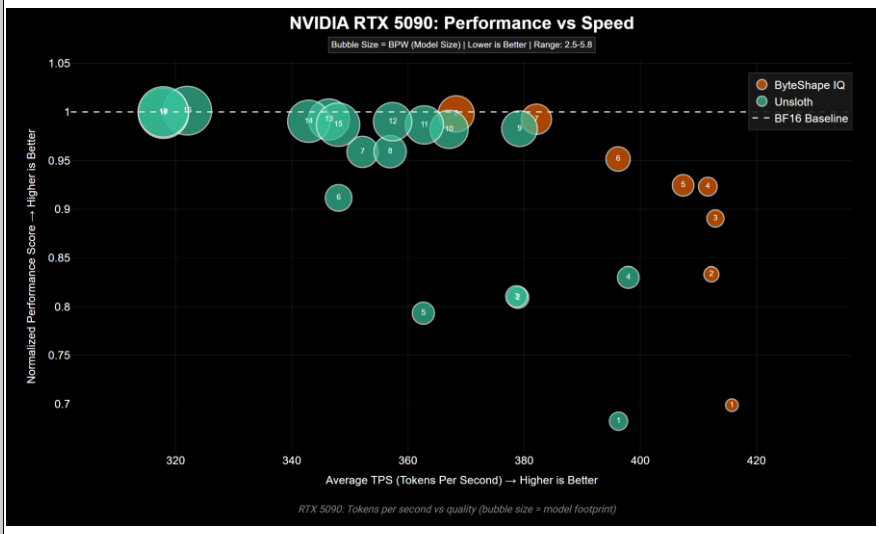


Real Performance: TPS (tokens/second)

*So far, we've looked at **model size vs quality**. In practice, speed is paramount to usability. We therefore measured throughput (TPS) for all variants on **NVIDIA GPUs**, **Intel CPUs**, and **Raspberry Pi**.*

GPU: RTX 5090

*The graph below shows **average TPS vs quality** for the **RTX 5090**, with bubble size proportional to model footprint. We again focus on models in the **2.7–5 bits/weight** range, which correspond to roughly **340–420 TPS** on the RTX 5090.*



*Our models cluster **to the right** (higher throughput) and **above** (higher quality) compared to the reference models. For example:*

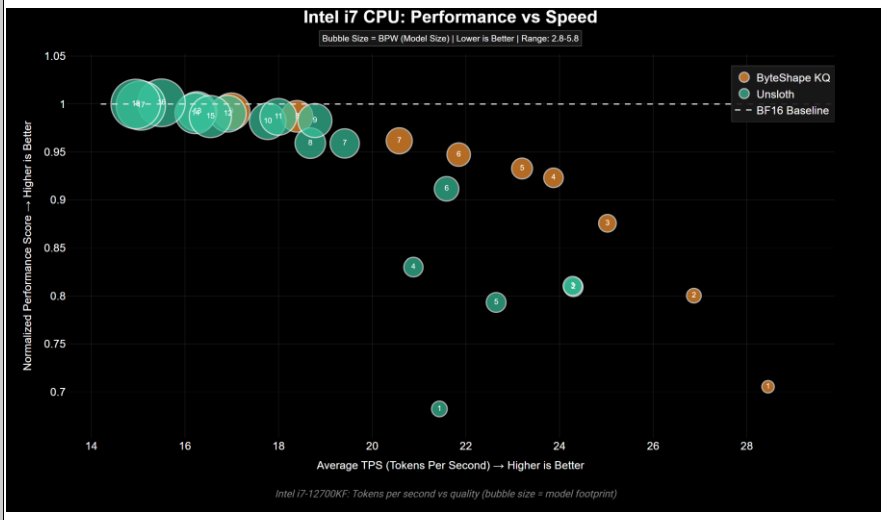
- *Our IQ-4.04bpw (382 TPS) [#7 on the graph] has both **higher throughput and higher quality** than all 13 Unsloth reference models in the **330–380 TPS** range. The first Unsloth model that surpasses it in quality is unsloth/Qwen3-4B-Instruct-2507-Q5_K_S [#16 on the graph] (322 TPS), which is **16% slower**.*
- *Our models in the top-right region (IQ-3.31bpw [#5 on the graph] and IQ-3.07bpw [#4 on the graph]) are alone at the top, combining **>90% accuracy** with the **highest throughput** among all evaluated models.*

*On GPU setup, low-bit **KQ** kernels are noticeably less efficient than **IQ** kernels. As a result, we constructed the GPU variants with a preference for IQ quantization.*

CPU: Intel i7 12th-Gen CPU

*The graph below shows **average TPS vs quality** for the i7-12700KF **CPU**, with bubble size proportional to model footprint. We again focus on models in the **2.7–5 bits/weight** range, which correspond to roughly **16–30 TPS** on the Intel i7 CPU. Contrary to the behaviour observed on our GPU setup, the KQ kernels outperform the IQ kernels on the Intel CPU. Accordingly, we suggest trying the KQ models first if you plan to run on*

CPUs.



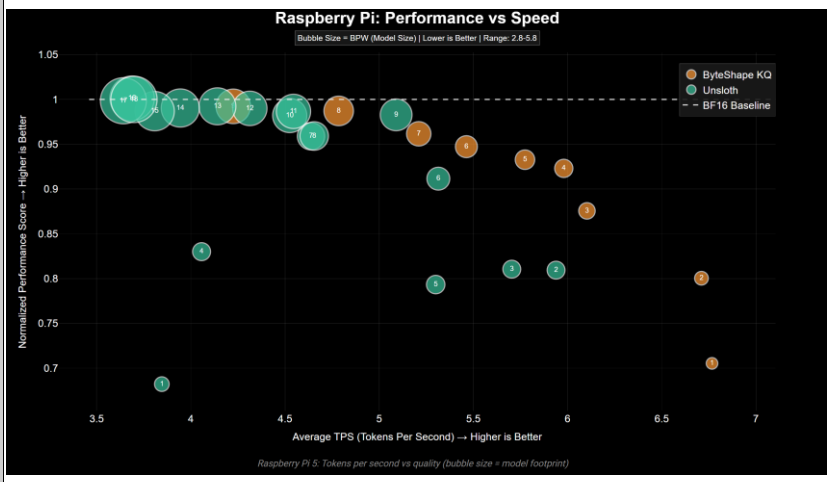
*Just like with the RTX 5090, our models cluster **to the right** (higher throughput) and **above** (higher quality) compared to the reference models.*

For example, our model KQ-3.34bpw [#4 on the graph]:

- *is more than **10% faster** than unsloth/Qwen3-4B-Instruct-2507-Q3_K_S [#6 on the graph] and noticeably **better quality** (1.15x smaller error rate),*
- *delivers 2.46x **smaller error rate** than unsloth/Qwen3-4B-Instruct-2507-Q2_K_L [#2 on the graph] with slightly smaller throughput (2% less TPS).*

Raspberry Pi 5

*Finally, the graph below shows **average TPS vs quality** for the **Raspberry Pi 5**, with bubble size proportional to model footprint. We again focus on models in the **2.7–5 bits/weight** range, which correspond to roughly **4–7 TPS**. Similar to the behaviour observed on our Intel i7 12th-Gen CPU, the KQ kernels outperform the IQ kernels.*



*Exactly like with the Intel CPU, our models cluster **to the right** (higher throughput) and **above** (higher quality) compared to the reference models on Raspberry Pi.*

The exact same model KQ-3.34bpw [#4 on the graph]:

- *is almost **13% faster** than unsloth/Qwen3-4B-Instruct-2507-Q3_K_S [#6 on the graph] and at noticeably **better quality** (1.15x smaller error rate),*
- *delivers 2.48x **smaller error rate** than unsloth/Qwen3-4B-Instruct-2507-Q2_K [#2 on the graph], this time with ever so slightly better throughput than the Unsloth reference (only 0.5% though).*

4. R&R (Roles and Responsibilities)

(max 10 pages)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* *We will be studying the potential of our technologies for AI model optimization on several use cases that would be of interest to potential Korean SMEs. We will delivering reports and models detailing our findings regarding the benefits delivered by the optimizations on the identified use cases.*
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* *We would appreciate the following contributions:* *1. Identification of relevant use cases for AI.* *2. Help with identifying relevant AI models and hardware/software platforms of interest.* *3. Help with refining metrics and benchmarks if of interest.* *4. Be available for regular consultations if possible.* *5. Be open to evaluating optimized models internally.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* *Tangible outcomes:* • *A set of models targeting several tasks of interest that have been optimized to meet performance, energy, real-time and cost constraints on hardware and software platforms of interest. When permissible the models will be released for public use.* • *Refinement of technologies for optimizing AI models and for compressing them.* • *Establishment of collaborations between UofT researchers and Toronto developers and Korean partners.* ***Commercialization Potential:*** *The PI and three of his former students have founded ByteShape Inc. which is commercializing the core technologies. ByteShape will be able to contribute to the project and will be able to commercialize the advancements in technologies and offer recurring services if needed by the Korean partners. The underlying technology IP has been developed at UofT by our group and is co-owned by UofT and ByteShape.*

6. References

- *BitPruning: Learning Bitlengths for Aggressive and Accurate Quantization*
- *Schrödinger's FP: Dynamic Adaptation of Floating-Point Containers for Deep Learning Training*
- *Every Bit Matters: A Hardware/Software Approach for Enabling More Powerful Machine Learning Models*
- *Learning Efficient Datatypes for Deep Neural Network Training and Inference*

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